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POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
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RESEARCH TITLE

**DEVELOPMENT OF ACADEMIC RESEARCH MONITORING SYSTEM
WITH DATA ANALYTICS FOR POLYTECHNIC UNIVERSITY OF THE
PHILIPPINES UNISAN, QUEZON CAMPUS**



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Introduction

Research is important because it helps solve problems and improve knowledge in many fields. As technology grows, Research Monitoring Systems and data analytics help organizations manage information, track progress, and make better decisions.

Many schools still use manual or outdated methods, which lead to delays, lost documents, and slow communication. This problem is also seen at PUP Unisan, where research monitoring is not centralized, making it hard for advisers and students to track progress properly.

This study aims to create and evaluate a web-based Research Monitoring System with Data Analytics for PUP Unisan. The system will help students and advisers monitor tasks more easily, improve communication, and ensure accurate and organized documentation. It also supports quality education and promotes efficient, transparent, and technology-based research processes.



Relevant Literature

This chapter delves deeply into what is already proven about research monitoring. It focuses on the existing literature and studies that have a direct correlation to the development of Research Monitoring System with Data Analytics. It examines the current trends, issues, and theoretical foundation relative to research monitoring and information systems. The chapter identifies gaps of previous works and establishes the basis for designing a system suited to the needs of PUP Unisan, Quezon Campus.



Relevant Literature

Review of Existing Research Monitoring System and Their Limitations

- This section of the study delves into the existing research monitoring systems and their possible gaps. Researchers will determine the existing systems' advantages and the recurring gaps that are worth bridging. This segment is about identifying the issues and understanding why the study is being conducted.**
- Research Monitoring System involves evaluating commercial and open-source platforms that help academic institutions, research libraries, and showcase research activities. System monitoring is an important basis for system modelling and improvement. Monitoring and evaluation are essential components of the research process, as they help maintain the quality, integrity, and timely completion of research outputs. Establishing an effective monitoring system is crucial for strengthening research governance (Enciso, 2024).**



Relevant Literature

Principles and Approaches in Designing Research Management System

- In this segment of the study, the focus is on the principles and approaches in designing the Research Monitoring System tailored for the Polytechnic University of the Philippines (PUP) Unisan, Quezon Campus. The researchers will determine which design approaches are appropriate to apply and what principles should guide the system's development. This section emphasizes the importance of establishing a well-structured and effective research monitoring system design.
- The Research Monitoring System is a prototype for HEIs in the Philippines to improve research management. It has a data input module (Fig. 1A) for recording application type, title, publication details, abstract, keywords, and research documents. A second module (Fig. 1B) stores author and co-author information and links research outputs to institutional goals. These modules support organized data handling, fast access, evaluation, and tracking. The system is designed to be user-friendly, adaptable, and helpful for decision-making (Regla & Ballera, 2024).



Relevant Literature

Integration of Data Analytics in Research Management

- This section explains how data analytics can be integrated into a research monitoring software system. By embedding analytics tools directly into the platform, the system can automatically process, analyze, and visualize research data. Through these features, the system can provide real-time insights, tracking research progress, detect irregularities, and generate reports that support data-driven decision-making.
- According to IBM (2024), embedded analytics helps improve system monitoring by offering real-time tracking and predictive insights that can warn users of delays or possible issues.



Relevant Literature

Software Evaluation of System Usability, Functionality and Reliability

- The modern era of organizational management and research is fundamentally characterized by the strategic deployment of data-driven systems for monitoring and decision support. This section examines how systems are evaluated in terms of their capabilities and accuracy based on ISO 25010 standards, particularly Usability, Functionality and Reliability. Through these criteria, a system's overall performance and quality are assessed.**
- Assessing software quality plays an important role in determining whether a system can effectively support user needs. The ISO/IEC 25010 standard provides a systematic way to measure software quality across several attributes, making it a reliable foundation for system evaluation (Efendi et al., 2025).**



Relevant Literature

Synthesis of the Reviewed Literature and Studies

- Existing research monitoring systems play a crucial role in managing and overseeing research activities within academic institutions. Systems like Research Information Management Systems (RIMS) and Current Research Information Systems (CRIS) provide centralized platforms for storing research metadata, supporting decision-making, resource allocation, and performance evaluation (Schöpfel & Azeroual, 2024; de Castro, 2020).



Methodology

This chapter represents the methodology used to carry out the study. It involves the research design, research locale, sources of data, research instrument and other procedures followed in developing and evaluating the Research Monitoring System with Data Analytics. This chapter also describes the ethical considerations observed throughout the research process.

RESEARCH DESIGN

- The study uses a Developmental Research Design, which involves the systematic process of designing, developing, and evaluating tools or models (Ramirez, 2023). This design fits the study because it aims to create an innovative system that addresses gaps in Information Technology.**
- The process includes analysis, design, development, implementation, and evaluation, ensuring the system is improved based on user needs and feedback. This approach also requires validation using both quantitative and qualitative methods.**



Methodology

RESEARCH LOCALE

- This study will be conducted at Polytechnic University of the Philippines (PUP) Unisan, Quezon Campus, which offers programs with thesis and capstone projects. The campus is suitable because it actively conducts research projects and provides accessible respondents in a relevant academic setting.
- Data will be collected from 3rd and 4th-year students in the Diploma and Bachelor of Science in Information Technology programs (total population = 138), along with research coordinators and faculty advisers.



Methodology

SOURCE OF DATA

- Primary data will be collected from the main users of the system—students, faculty members, and the Research Coordinator at PUP Unisan. They can provide insights on user needs, current problems with research monitoring, and feedback on the system’s usability and functionality. Data will be gathered through structured surveys for quantitative measures like satisfaction and performance, and interviews for qualitative insights, including expectations, challenges, and system requirements.
- Secondary data will come from official university documents, reports, research manuals, and system logs. These sources provide context, operational data, and historical records to support analysis of the current research monitoring process and help develop the data analytics module.



Methodology

RESEARCH INSTRUMENT

- This study will use questionnaires and surveys to collect data from students and advisers about the Research Monitoring System. The questionnaire, adapted from Urera Jr. and Balahadia (2019), uses a 5-point Likert Scale to measure perceptions, experiences, and satisfaction. Surveys will provide additional insights using the same structured format for consistency.
- All data collected will be kept confidential, and respondents' identities will not be disclosed.



Methodology

DATA GATHERING PROCEDURES

- Data will be collected from students in the Diploma and Bachelor of Science in Information Technology programs at PUP Unisan who are currently engaged in research activities. The primary instrument is a questionnaire, complemented by a survey for additional insights, both aligned with the study's objectives and SDGs.
- The study will begin with permission from the Campus Director, followed by coordination with the research coordinator and advisers to schedule data collection. Questionnaires will be distributed with clear instructions and study objectives, collected after completion, and systematically coded for analysis. Participants will provide informed consent, and all responses will be kept confidential.



Methodology

STATISTICAL DATA ANALYSIS

- This study uses descriptive statistics to analyze data from students and advisers collected through a five-point Likert scale questionnaire on the system's usability, functionality, and reliability. Frequency counts and percentages summarize demographics, while weighted means and standard deviations assess agreement levels and response consistency.

- Data were checked for completeness, encoded, and processed using Excel. Findings are presented in tables and narratives, providing an objective overview of user perceptions and evaluating whether the system meets quality standards. Inferential statistics may be used if needed, but descriptive analysis forms the main basis for evaluating the Research Monitoring System.



Methodology

ETHICAL CONSIDERATIONS

- The study followed strict ethical guidelines to protect participants' rights, privacy, and safety at PUP Unisan. The purpose, methods, risks, and benefits were clearly explained, and all participants gave informed consent. No personally identifiable information was collected; responses were coded, kept confidential, and securely stored. Participants could withdraw at any time without penalty.
- For system testing, only anonymized data were used, and all research data followed the Data Privacy Act of 2012 with encryption and access controls. Analytics features were designed to avoid bias and ensure accuracy. Required approvals were obtained from UREC, research advisers, coordinators, and campus administration. Ethical standards and laws were strictly followed throughout the study.



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&text=Highly%20sensitive%20requires%20constant%20monitoring,inadequate%20expansion%20capability.](https://www.scribd.com/document/100487524/Disadvantages-of-Management-Information-System#:~:text=Institutionalisation%20of%20the%20system%20%2D%20Has%20it%20been%20taken%20on%20board%20enthusiastically?&text=Highly%20sensitive%20requires%20constant%20monitoring,inadequate%20expansion%20capability.)**



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